Radar Tracking Software Pipeline Optimisation for UAV Detect-and-Avoid

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B.Eng (Hons)

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the requirements for the degree of
Bachelor of Engineering (Honours)

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Abstract

TODO: Radar is able to provide Unmanned Aeriel Vehicales with the capability to detect and track airborne hazzards over large distances and in all weather conditions. Talk about trade off between size, power, weight and compute. Talk about problems.

TODO: We seek to address these problems through adapting establish radar processing algorithms for UAVs. By evaluating the effectiveness of different implementations and algorithms essential to multi target tracking, this report demonstrates how software choises make long range radar practical for small UAV detect and avoid systems.

TODO: Talk about key parts of our pipeline

TODO: Talk about the Experiments

TODO: Talk about findings

TODO: Ultimately, our work provides a reference for engineers and policy makers...

Statement of Contributions

I, **Noah West**, declare that the intellectual content of this thesis is the product of my own work and that all assistance received in the preparation of this thesis and relevant sources have been appropriately acknowledged. The contributions I made to this research include:

- Undertaking the background research and completing the literature review.
- Defining the research scope and developing the problem statement in consultation with my supervisor and the industry partner.
- Designing the methodology, experimental approach, and test plan with guidance from my supervisor.
- Designing and implementing, in collaboration with engineers from the industry partner *Mission Systems*, the software pipeline described in this work.
- Profiling and evaluating the pipeline software to produce the reported results.
- Capturing radar data for use in testing and validation.
- Conducting the experiments, including configuration of the test setup and execution of data collection procedures.
- Performing analysis of results, including quantitative evaluation using appropriate metrics to validate conclusions.
- Interpreting findings, and writing the discussion and conclusions, incorporating feedback from my supervisor and relevant stakeholders.
- Writing the thesis and preparing figures, tables, and supporting material.

The above represents an accurate summary of my contribution.

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Acknowledgements

TODO: Make it a proper acknowledgements section

Thank you to my adoring fans. Thanks to my supervisor Dr. Donald Dansereau for being the goat.

I would like to extend thanks Rex for being the goat.

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CHAPTER 1

Introduction

1.1 Motivation and Context

With more uncrewed aerial vehicles (UAVs) in the air each year, and a growing desire for them to operate in controlled airspaces, there is a concern regarding their ability to maintain separation from hazards in the air. United States and European federal agencies are developing policy that necessitates detect-and-avoid (DAA) systems for UAVs, which parallel the visual and instrument flight rules (VFR & IFR) that dictate how pilots avoid traffic [1] [2] [3].

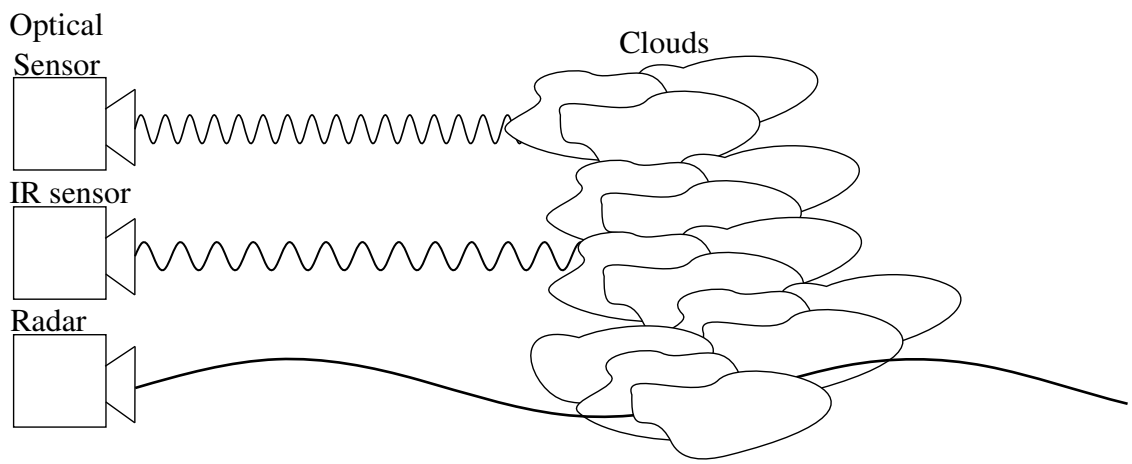

FIGURE 1.1: Just a test diagram

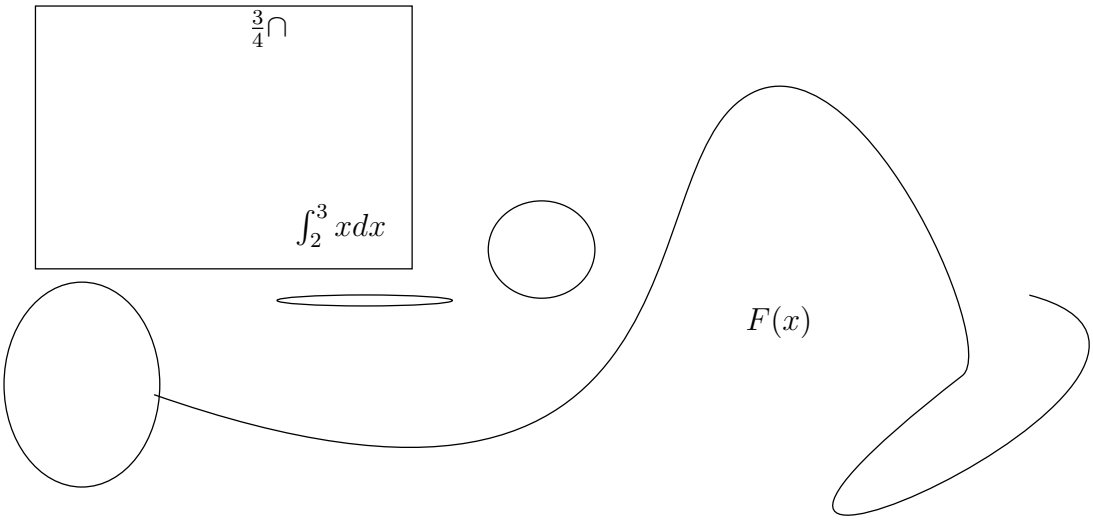

FIGURE 1.2: Abstract Architecture

FIGURE 1.3: A monopulse radar

While some proposed DAA systems utilise optical and infrared sensors [3], radar sensors may also be suitable, as they can perform sensing over great distances in all weather conditions, due to their relatively low atmospheric attenuation (compared with optical systems) [4]. The inconvenient side effect of their long wavelength property is that target identification by radar

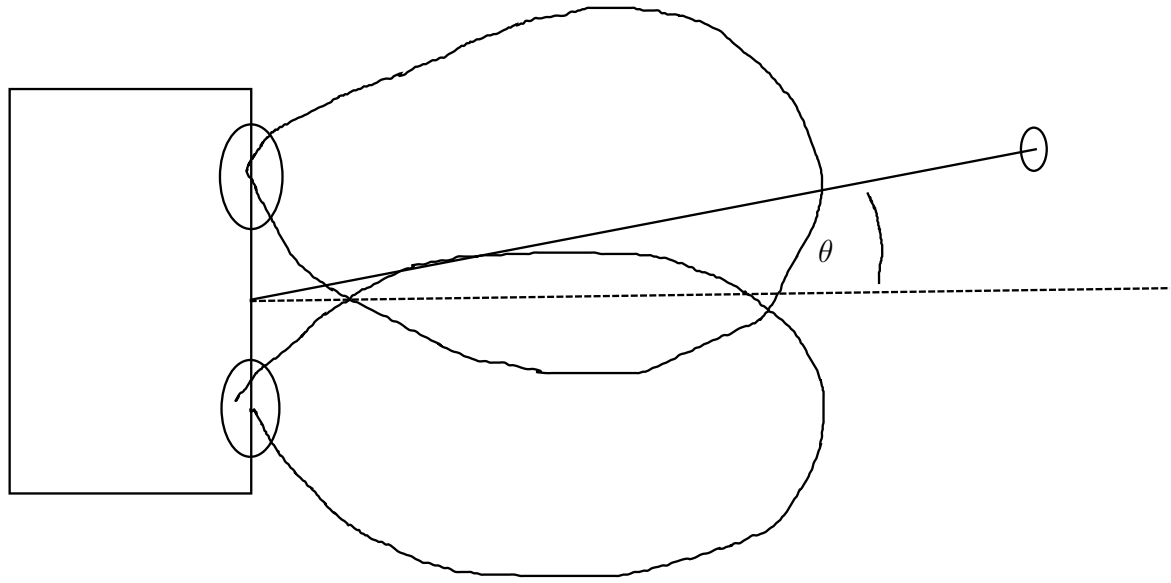

FIGURE 1.4: A diagram of a monopulse radar

systems is less effective than imaging systems that use shorter wavelengths, such as optical trackers. If this challenge is overcome, the implementation of radar as the primary sensing mode or an additional sensing mode will improve the awareness, and thus the safety of UAVs.

In contrast to larger crewed aircraft that can afford great computational resources and large aperture radar, radar systems fit for the size, weight and power (SWaP) conditions of small to medium UAVs may encounter constraints such as low processing power, and a poor signal-to-noise ratio (SNR). Despite the less than ideal conditions for radar on UAVs, they are the fastest growing market for radar systems, projected to grow with a compound annual growth rate (CAGR) of 11.22% globally through 2030 [5]. To make these emerging radar systems more performant, accessible, and safe, it would be valuable to find the most suitable radar signal processing pipeline for UAV DAA systems, revisiting the old problem of radar target tracking in the face of unique hardware conditions.

Software-defined architectures are increasingly preferred over hardware defined radar systems for UAV applications as software modifications can be shipped faster than hardware modifications [5]. As a result, graphics processing units (GPUs) are favoured over non-programmable

digital signal processors such as application specific integrated circuits (ASICs). GPU-accelerated implementations are also favourable for computationally intensive stages of the radar processing chain, as many core operations, such as fast Fourier transforms (FFTs) and matrix operations, are highly data-parallel and can leverage mature GPU software ecosystems and optimised libraries [6]. Additionally, the incorporation of AI into radar tracking algorithms has further encouraged these highly parallel, GPU-based pipelines [7]. Key open questions remain around balancing the compute saved through GPU parallelism against the cost of data movement and synchronisation, and how unified (shared) CPU–GPU memory on embedded platforms changes this trade-off.

TODO: Write industry context section heading/intro. Most detail will go in background section so just give high level

TODO: Current UAV Laws. Australia, USA, Europe, Canada. DAA and how its necessary for UAVs to be integrated into public airspaces.

TODO: What size vehicles are currently given radar systems. If small UAVs have them, how does the size of the UAV affect the quality/performance of the system.

1.2 Knowledge Gap

There are two main adverse effects of radar receivers that are sized for small to medium UAVs. The first is low received power, which is proportional to the effective receive aperture area. Antenna gain G scales with effective aperture A_e as $G \propto A_e$. All else equal, reduced gain lowers received power and therefore reduces SNR, with $\mathrm{SNR} \propto G^2$, decreasing the sensitivity of the system to small or far away targets. The second adverse effect is poor angular resolution. Beamwidth θ_{BW} scales as $\theta_{\mathrm{BW}} \propto 1/D$, where D is the effective aperture diameter. With a small radar system and thus a small effective diameter, beamwidth is large and resolution

is poor. Therefore, it becomes more difficult to resolve the velocity and positions of single targets, differentiate between multiple targets, identify their size, and have confidence in detections.

TODO: Talk broadly about the state of the current literature and how they engage with this new issue. Summarise the major gaps in the literature in terms of radar systems under swap constraints, and if the literature has missed any techniques that we're going to employ. Tie that into our aims and outcomes, explaining how our experiments will fill the gaps.

1.3 Problem Definition

While radar systems do provide advantages over optical and infrared sensors for detecting airborne targets, several challenges hinder widespread adoption by smaller UAVs.

TODO: List the largest barriers. What we will try and solve. Include SWaP-C constraints as one of those challenges. The complexity of these systems is itself a barrier to widespread adoption and through demonstrating the design of such a system and its capabilities, we lower this barrier for other engineers.

TODO: The field lacks a replicatable model for building and testing small radar DAA systems.

TODO: Figure out how the problem definition differs from the motivation.

1.4 Scope and Assumptions

TODO: Discuss Scope and Assuptions

TODO: Scope: Profiling of inidividual algorithm, profiling of integrated pipeline, sensitivity analysis. Cost analysis in terms of latency, memory, sensitivity. Partial field testing. Experimenting with different hardware restrictions.

TODO: Not in scope: Hardware Design, Full Field Testing. FPGA/ASIC based solutions. Comparing optical or infrared sensitivity to radar.

TODO: Assumptions: Conformal Monopulse Radar, GPU enabled hardware. Shared Memory

1.5 Contributions

TODO: Write aims and outcomes section heading/intro.

The proposed thesis will be a systems report regarding the software design space of a GPU-based radar tracking pipeline for a small monopulse radar. The problems of poor resolution and low signal strength described above will be addressed in the proposed project, a systems report that compares different signal processing and tracking algorithms for their ability to detect faint targets from small UAV platforms. Due to the growing interest in the hardware for UAV applications, these algorithms will be implemented on GPUs and their throughput and latency will be measured. Questions about the cost of memory movement and synchronisation will be addressed through integration testing and profiling of the entire pipeline. A portion

of the research will also contribute to an understanding of the effect that different hardware contexts on the optimal design of these radar processing pipelines.

In summary, the project will contribute the following:

- A comparison of candidate detection, angle-estimation, and tracking algorithm combinations, assessed by detection sensitivity and track quality in multi-target aerial scenarios, to inform algorithm selection.
- A comparison of alternative low-level implementations of the selected algorithms, quantifying their effects on end-to-end throughput and latency in highly parallelised GPU pipelines, to inform implementation choices.
- An assessment of how these algorithm and implementation conclusions vary across embedded hardware configurations, providing hardware-aware design recommendations.

TODO: Add any missing aims/outcomes content that is not present in the provided text.

TODO: Tie the aims and outcomes into the motivation

1.6 Thesis Structure

Chapter 2 introduces the background information related to the following areas: standard radar algorithms, monopulse angle estimation, multi-target tracking, and parallel processing.

Chapter 3 conducts and extensive literature review of both foundational and state of the art approaches to performant radar systems under high SWaP constraints. The methodologies of works with similar motivations to ours are critically evaluated for their effectiveness in meeting their aims. Key findings and gaps across works are summarised and an explanation for how this thesis fits in with the existing literature is provided.

Chapter 4 Explains the methodology by which we compared different radar processing pipeline designs for their suitability for use in detect and avoid systems, and compare different algorithms for their ability to detect small, faraway targets. This methodology will describe the algorithms, the implementations of these algorithms, the processes by which the implementations are profiled, and the processes by which the algorithms are compared for their impact on the overall sensitivity of the system.

Chapter 5 Presents the key findings from the experiments outlined in the methodology.

TODO: Continue this structure section for chapters 5, 6, 7

CHAPTER 2

Background

2.1 Aviation Policy

TODO: Discuss the state of policy across Australia, USA, Canada, Europe. Which contries share policy and which ones are leading the adoption of UAVs.

TODO: Discuss the current requirements for DAA systems. Describe what it would take for them to meet the standards of instrument or visual flight rules.

TODO: Is this the most appropriate place for policy to be discussed? It's important for motivation and goals but it isn't relevant to understanding the technology.

2.2 Monopulse Radar Hardware

TODO: Principles and methods of monopulse radar. Angle discrimination

TODO: Go into phased array antennas

TODO: Discuss Hardware Choices (out of scope)

TODO: Expected Accuracy, error, noises, simulation

2.3 Radar Processing

TODO: Talk about radar hardware and software fundamentals

TODO: How radar works (brief)

TODO: The typical stages, matched filtering vs mixing and fft, role of spectral domain. Ranging. Coherent integration and slow trasform to measure Doppler.

2.4 Multi Target Tracking

TODO: Filtering, Kalman Theory

TODO: Coordinate Systems (polar for radar)

TODO: Gating and data ascociation

$$f(x) = \frac{3}{2}$$

TODO: Track Life Management

2.5 Parallel Processing

TODO: Roles of CPUs, GPUs, ASICs, and other architectures in radar pipelines

TODO: Architecture of GPUs and how its lends itself to high performance for specific operations

TODO: The roles of L2 cache, branch predicability, warp and block sizes on speed of GPU based algorithms

TODO: Maybe a bit on how we utilise this understand to inform our experiment?

CHAPTER 3

Literature Review

3.1 UAV Radar System Feasibility

TODO: Develop these to more completely engage with what they're doing. Read the works fully

Sysak et al. [8] present a lightweight X-band monopulse radar designed explicitly for mid-air collision avoidance on medium UAV platforms, addressing the range shortfall that often limits compact frequency modulated continuous wave (FMCW)-based solutions. Their system integrates a multi-channel, forward-looking antenna with an FPGA-based pulse compression on a single 60 gram PCB, enabling online range processing and adaptive CA-CFAR detection, while keeping average power consumption below 80 Watts. Field trials in urban environments demonstrate reliable detection of large obstacles at 1–3 km, providing manoeuvring standoff times on the order of tens of seconds. This work is strongly aligned with UAV Detect-and-Avoid motivations and demonstrates that high-performance sensing can be achieved under strict SWaP constraints with a pulsed, multi-channel architecture.

However, the evaluation is primarily conducted against large, high radar cross-section infrastructure targets, and does not characterise sensitivity to small airborne targets in realistic encounter geometries. In addition, while the work acts as a proof of concept for UAV radar systems, it does not address multi-target tracking nor the latency, throughput and memory usage of a complete onboard pipeline. These omissions are important for DAA, where hardware is limited, and efficient data handling is as critical as raw detection range. The proposed

project complements Sysak et al. by benchmarking detection and tracking sensitivity on small moving targets, and by investigating GPU-oriented implementations and pipeline architectures suitable for low-compute embedded platforms, including the performance impact of memory movement.

3.1.1 End-to-end UAV DAA onboard processing pipelines

Feasibility studies with more emphasis on the radar DAA processing pipelines have been documented. Three that are especially notable include the systems designed by Shi et al. [9], Moses et al. [10], and Newmeyer et al. [11]. Shi et al, similarly motivated by the need to detect-and-avoid non-cooperative targets, describe the design of a small FMCW phased array built mainly from off-the-shelf development boards. The design is only suitable for large UAVs proposed project will target smaller platforms. Additionally, the full radar processing was not entirely onboard and online, splitting steps between an onboard FPGA and offboard post processing. On the other hand, the proposed research will design the software for fully onboard, online tracking.

Moses et al. illustrate the capabilities of a lightweight, single channel FMCW radar. Testing was performed on a quadcopter. A major component of their analysis emphasises identification via micro-Doppler modulation, which is largely evaluated through Doppler-domain signatures over end-to-end range–Doppler detection and tracking. While this is well motivated for classification, it doesn't address the tracking targets in the range–Doppler space. It also remains unclear how the small aperture and SWaP limits affect range-domain SNR, achievable range resolution, and the resulting false alarm behaviour. The proposed research fills this gap by benchmarking range–Doppler processing, CFAR detection, monopulse angle estimation, and multi-target tracking in low-SNR regimes, and by quantifying how these choices trade sensitivity against latency and throughput on embedded GPU hardware.

Finally, Newmeyer et al. demonstrated a 120 gram, 8 Watt FMCW phased array radar and processing pipeline that utilised the FPGA fabric on a Zynq ARM CPU. Power, compute cost, latency distributions, and hardware resources were closely monitored, and the project clearly

demonstrates that radar DAA systems can operate under strict SWaP constraints. However, the study only evaluates one hardware configuration, and one radar algorithm sequence. The proposed research fills this gap by systematically comparing alternative algorithmic pathways and low-level implementations, and by evaluating how pipeline bottlenecks and end-to-end performance change across different embedded compute and memory configurations, including GPU-based architectures.

While these works demonstrate early feasibility of small-platform DAA radar, their hardware stacks are not fully representative of modern embedded radar processing platforms in the sense that they do not utilise GPUs or occur entirely onboard, and their evaluations do not explore how algorithmic selection relates to sensitivity to faint target signals. Consequently, there remains a gap in the literature regarding how to integrate and benchmark alternative radar processing and tracking algorithms for small conformal apertures under SWaP constraints, particularly when optimising jointly for sensitivity, latency, and throughput.

3.2 GPU Radar Processing Performance

TODO: Maybe write about other papers that broadly explore GPU acceleration in the context of any sensors or in the UAV context

3.2.1 GPU-accelerated radar signal processing

TODO: Develop this futher

Studies that are less UAV focused, such as the GPU-accelerated passive bistatic radar pipeline of Zhao et al. [12] and the real-time GPU radar processing thesis of Zaknoun [13] provide more comprehensive analyses of radar software architecture and GPU implementation considerations. Zhao et al. implement compute unified device architecture (CUDA) acceleration

for key stages, reporting up to a 113.13× speedup over a CPU baseline. The research is limited to only three algorithms (ECA-B clutter suppression, range–Doppler processing, and 2D CA-CFAR detection), leaving gaps for similar research that applies to other common radar algorithms such as angle sensing, coordinate transforms, kalman filters, and more. Additionally, the research did not explicitly discuss the time cost of memory reallocation and thread synchronisation. The proposed project will fill these gaps from this study by benchmarking a more comprehensive selection of tracking algorithms for CUDA implementations, and profiling bottlenecks such as host–device data transfer and branching delay to provide a clearer understanding of the benefits of parallelised processing.

3.3 Low SNR Mono-pulse Feasibility

TODO: This is an opportunity to bring in really established papers on monopulse radar and explain what they do to maximise SNR. It may be hard to find gaps that aren't obvious.

TODO: Develop the below further

In addition to a lower SNR, a key limitation of SWaP-constrained monopulse radars is that a reduced aperture degrades angular resolution, increasing the likelihood that multiple targets occupy the same angular cell and that angle estimates begin reflecting the centroid of the target group. An et al. [14] propose an efficient method for resolving two unresolved targets in a monopulse cell, but evaluation is limited to simulated white-noise conditions and does not address integration into an end-to-end range–Doppler detection. The method also has explicit failure modes (same azimuth/elevation; three or more targets in-cell). To address these gaps, our research will test this algorithm with real simulation data, where noise is not necessarily white, and compare how the incorporation of the algorithm into a tracking system improves track quality.

Complementing this, Aubry et al. [15] demonstrate that detection and angle estimation can be treated jointly in a multichannel array, rather than as a strict two-stage pipeline (detect then estimate), which is attractive under SWaP conditions where false alarms and low SNR make early thresholding brittle. The research demonstrates an algorithm that fuses data between channels before angle tracking occurs, increasing sensitivity to targets that are not visible in any single channel. The research exclusively uses single simulated targets in Gaussian interference to validate the algorithm, leaving a gap in characterising the performance under real data. The proposed project will address these gaps by applying the algorithm to alternate noise regimes and multiple target scenarios.

3.4 Programmable Vision Acceleration for Radar

TODO: Find works that explore NVIDIA's PVA in the context of Radar

3.5 Summary

3.5.0.1 End-to-end systems level comparisons under SWaP constraints

Across the UAV DAA radar literature, feasibility has been demonstrated, but evidence remains fragmented across hardware prototypes, single-path processing demonstrations, and isolated GPU acceleration studies. As a result, there remains a gap in consolidated, systems-level research that illustrates which combinations of radar signal processing, monopulse angle estimation, and multi-target tracking algorithms are most effective when jointly optimised for sensitivity, latency, and throughput on GPU-based embedded platforms, and how these conclusions change with hardware configuration. The proposed evaluation of small-target sensitivity and end-to-end tracking on embedded GPU hardware directly supports our contributions on algorithm selection, low-level implementation trade-offs, and hardware-aware design recommendations.

3.5.0.2 Realistic small-target and encounter-focused evaluation

The experimental design explicitly includes small moving aerial targets and encounter-like scenarios in both simulation and real data collection. Performance is measured not only by detection range but by false alarm behaviour, track continuity, and sensitivity in low-SNR regimes that better reflect small-aperture UAV radars. By extending prior single-path demonstrations into a comparative study of alternative algorithm combinations and fully onboard GPU implementations, this section motivates our contributions on algorithm selection and implementation choices, and on how these choices vary across hardware contexts.

3.5.0.3 GPU-oriented onboard pipeline evidence, including data movement costs

The thesis targets embedded GPU hardware and evaluates architectural choices that often dominate real-time performance, including host–device transfers, memory layout, streaming formats, kernel fusion opportunities, and synchronisation overhead. This thesis advances the literature by benchmarking low-level GPU implementations beyond range–Doppler and CFAR, quantifying end-to-end latency and throughput impacts, and assessing how these conclusions change across embedded hardware configurations.

3.5.0.4 Hardware-aware design conclusions across embedded configurations

Experiments are repeated across multiple hardware contexts (for example CPU-only, unified-memory embedded GPU, and discrete CPU+GPU settings where available) and across constrained power and clock modes. Testing these angle-estimation methods within a complete multi-target tracking pipeline, and measuring both sensitivity and track-quality impacts alongside compute cost, directly supports our contributions on algorithm selection, implementation trade-offs, and hardware-aware recommendations.

CHAPTER 4

Methodology

TODO: Describe the design of our experiment. Break down experiment into sub experiments, how does that relate to the different application we are building.

TODO: Describe the tools and processes we will use to validate our software. Nsight systems, nsight compute, other tools for measuring memory. Bash scripts for testing

TODO: Below is how we test things. Then after, we will discuss what we're testing.

4.1 Testing Datasets

TODO: Discuss the different sources of data. Garbage data / sine waves, Ocean Acoustics library data, false target data from AWR chips and real target data from AWR chips

TODO: What insights will we get from testing with each of these datasets, and what are the challenges ascociated from use the more sophisticated ones

4.2 Performance Analysis

4.2.1 Throughput Analysis

TODO: Measure in Bytes Per Second

4.2.2 Latency Analysis

TODO: Use NSight

TODO: Ablation Study

TODO: Explore the effect of different hardware contexts on the performance. What contexts prevent us from implementing certain solutions and why

4.3 Tracking Performance Analysis

TODO: Ablation Study

TODO: Try different alternatives for each operation

TODO: The below sections details what we're testing

4.4 Processing Application

TODO: Complete the following sections. How do the experiments resolve the gaps in the literature.

4.4.1 Data Input

4.4.2 Shaping Data

4.4.3 Windowing

4.4.4 Fast Fourier Transforms

4.4.5 Coherent Integration

4.5 Tracking Application

TODO: Complete the following sections. How do the experiments resolve the gaps in the literature.

4.5.1 Constant False Alarm Rate

4.5.2 Monopulse Angle Sensing

4.5.3 Embedding

4.5.4 Polar Transform

4.5.5 Data Ascociation

4.5.6 Tracking Filters

4.5.7 Initiating and Deleting

4.6 Orchestration and Dataflow

4.6.1 Threads and Batching

4.6.2 Reading data

4.6.3 Writing data

4.6.4 Aggregating Data Streams

4.7 Results

TODO: Mirror the methodology

4.8 Orchestration and Dataflow

4.8.1 Threads and Batching

4.8.2 Reading data

4.8.3 Writing data

4.8.4 Aggregating Data Streams

4.9 Processing Application

4.9.1 Data Input

4.9.2 Shaping Data

4.9.3 Windowing

4.9.4 Fast Fourier Transforms

4.9.5 Coherent Integration

4.10 Tracking Application

4.10.1 Constant False Alarm Rate

4.10.2 Monopulse Angle Sensing

4.10.3 Embedding

4.10.4 Polar Transform

4.10.5 Data Association

4.10.6 Tracking Filters

CHAPTER 5

Discussion

5.1 Orchestration and Dataflow

TODO: Identify the best strategy for each of the subsections. In terms of our different performance metrics.

TODO: What do our tools and result suggest is happening on a low level. Come up with explanations for any phenomina which isn't immediately obvious (i.e. most things)

TODO: How do these findings relate to the results to the other sections. Can we fuse any operations together or take any cleaver short cuts

TODO: What hardware allows us to do which options, i.e. Direct Memory Access, Shared Memory

TODO: Explicitely address the knowledge gaps and contributions

5.1.1 Threads and Batching

5.1.2 Reading data

5.1.3 Writing data

5.1.4 Aggregating Data Streams

5.2 Processing Application

TODO: Discuss the results for each of the operations. relate to our contributions, gaps and results from the other areas of research.

5.3 Tracking Application

TODO: Discuss the results for each of those. relate to our contributions, gaps and results from the other areas of research.

5.4 Real World Implications

TODO: Discuss how our findings either support or don't support the feasibility of radar systems for small UAVs. What size UAVs does this work for? How should policy be written based on our findings. How should people feel about the safety.

TODO: Cultural, Social, Policy, Environmental Implications. Ethics

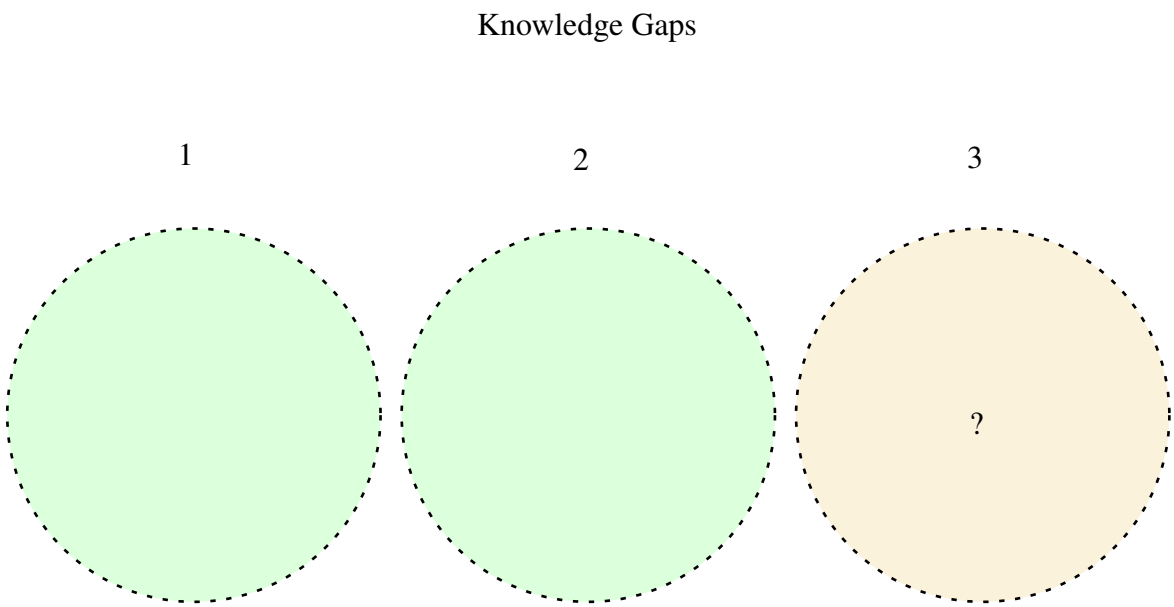

FIGURE 5.1: Me just messing around with Inkscape

TODO: Recommendations for Engineers.

5.5 Addressing the Identified Knowledge Gaps

TODO: Explicitely address the knowledge gaps

CHAPTER 6

Conclusion

6.1 Summary of Findings

TODO: Summarise the finds at a high level. State the best/recommended configuration.

TODO: Link together all the sections of the discussion into main takaways. Kinda like an executive summary.

6.2 Industry Partner Outcomes

TODO: Without sharing anything sensitive, detail how the research done benefits mission systems in their project goals.

TODO: Document tests and downstream work that my work enabled.

TODO: Discuss how this technology will after the industry outside of Mission Systems too.

6.3 Future Work

TODO: What about our research was successful, what wasn't successful

TODO: Discuss the limitations of the set of computing hardware that we were able to explore

TODO: We only explored one radar device

TODO: Talk about algorithms that were not explored

TODO: Compare to other sensing systems

TODO: Sensor Fusion

Bibliography

[1] European Union Aviation Safety Agency (EASA). (Jul. 2024). Easy access rules for unmanned aircraft systems (regulations (eu) 2019/947 and 2019/945): Revision from july 2024. Online publication page (Revision from July 2024), (visited on 04/02/2026).

[2] RTCA, Inc. (Dec. 2025). Sc-228: Minimum operational performance standards for unmanned aircraft systems. Committee page; Terms of Reference listed as December 16, 2025, (visited on 04/02/2026).

[3] C. Serres, B. Gill, M. W. M. Edwards and M. G. Wu, 'Detect-and-avoid safety and operational suitability analysis using an electro-optical/infrared sensor model,' National Aeronautics and Space Administration, NASA Technical Memorandum NASA/TM–20210016872, Mar. 2021, Ames Research Center; MIT Lincoln Laboratory.

[4] B. Borden, *Radar Imaging of Airborne Targets: A Primer for Applied Mathematicians and Physicists*. CRC Press, 1999, Originally published 1999; later reprints available via CRC Press/Taylor & Francis.

[5] Mordor Intelligence. (Jul. 2025). Airborne radars market size & analysis - growth trends & forecasts (2025 - 2030). Page last updated on July 24, 2025, (visited on 04/02/2026).

[6] K. Zaknoun, 'Real-time, gpu-accelerated processing of digital radar signal data,' Master's thesis, University of Turku, Jun. 2024.

[7] W. Feng, X. Hu and X. He, 'Artificial intelligence (ai)-based radar signal processing and radar imaging,' *Electronics*, vol. 13, no. 21, 2024.

[8] D. Sysak, A. Byndas, T. Karas and G. Jaromi, 'Lightweight and compact pulse radar for uav platforms for mid-air collision avoidance,' *Sensors (Basel)*, vol. 25, no. 23, p. 7392, Dec. 2025.

[9] L. Shi, C. Allen, M. Ewing, S. Keshmiri, M. Zakharov, F. Florencio, N. Niakan and R. Knight, 'Multichannel sense-and-avoid radar for small uavs,' in *2013 IEEE/AIAA 32nd Digital Avionics Systems Conference (DASC)*, 2013, 6E2-1-6E2–10.

[10] A. Moses, M. J. Rutherford and K. P. Valavanis, ‘Radar-based detection and identification for miniature air vehicles,’ in *2011 IEEE International Conference on Control Applications (CCA)*, 2011, pp. 933–940.

[11] L. Newmeyer, D. Wilde, B. Nelson and M. Wirthlin, ‘Efficient processing of phased array radar in sense and avoid application using heterogeneous computing,’ in *2016 26th International Conference on Field Programmable Logic and Applications (FPL)*, 2016, pp. 1–8.

[12] X. Zhao, P. Liu, B. Wang and Y. Jin, ‘Gpu-accelerated signal processing for passive bistatic radar,’ *Remote Sensing*, vol. 15, no. 22, 2023.

[13] K. Zaknoun, ‘Real-time, gpu-accelerated processing of digital radar signal data,’ Master of Science in Technology Thesis, University of Turku, Turku, Finland, 2024.

[14] Q. An, C. Yeh, Y. Lu, Y. He and J. Yang, ‘Time-varying angle estimation of multiple unresolved extended targets via monopulse radar,’ *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 4, pp. 4650–4667, 2024.

[15] A. Aubry, A. D. Maio, S. Marano and M. Rosamilia, ‘Single-pulse simultaneous target detection and angle estimation in a multichannel phased array radar,’ *IEEE Transactions on Signal Processing*, vol. 68, pp. 6649–6664, 2020.

APPENDIX A

Appendix

A1 Case Studies

A2 Case Studies: Learning Objectives Covered by Thesis

For ESIPS substitution, this thesis is proposed to replace content from two elective units: **AMME5520 Advanced Control and Optimisation** and **ELEC3305 Digital Signal Processing**. The proposed thesis covers over 75% of learning objectives from each unit outline through the necessary self-guided learning and engineering design required to meet the contributions.

A2.1 Case Study 1: AMME5520 Advanced Control and Optimisation

A2.1.1 LO1. *Approach research papers in a professional and research-orientated manner, and conduct critical reviews of these papers*

This learning objective is met through the thesis literature review and gap synthesis across UAV DAA radar pipelines, GPU radar processing, monopulse angle estimation under low SNR, and multi-target tracking. The thesis requires evaluating assumptions, identifying limitations, and justifying design choices based on evidence, aligning with Week 13 Lightning talks (LO1) and the unit's self-directed critical engagement with research.

A2.1.2 LO2. *Implement simple path generation algorithms, controllers and decision metrics for an autonomous system in order to meet specific mission objectives*

This learning objective is met by implementing autonomy-relevant estimation and decision components in the detection-to-track chain: measurement formation, data association (e.g., NN/GNN/JPDA variants), tracking filters (EKF/UKF/IMM-style variants), and track lifetime management. Although the thesis project does not explicitly control the position of the targets that it is making estimations on, estimation is an essential component of any control system, and multi target tracking engages with the state based decision making that this learning objective is concerned with. The estimation theory in this project has heavy alignment with Weeks 07–08 in particular (Kalman filtering and extensions).

A2.1.3 LO3. *Understand a number of different path generation and control algorithms implemented in autonomous systems and how they are linked to optimality criteria, platform stability and vehicle constraints.*

Tracking aerial vehicles that can both fly in a straight line and conduct several manoeuvres motivates the use of multiple filters and tracks that model possible trajectories of the targets. Although targets are not being controlled, multi target tracking itself is an optimisation problem. Understanding stability is essential to minimise track id switching when multiple targets are within a few standard deviations of one another (in terms of state).

This learning objective is also met through the comparative evaluation of tracking algorithms, where we optimise for high track quality and low computational resource use. This aligns with Weeks 02–03 (optimality-driven algorithms), Weeks 07–09 (estimation, uncertainty, robustness).

A2.1.4 LO4. *Understand how cost functions are used to define mission objectives in a mathematical form, so that autonomous systems can make decisions about their next action.*

This learning objective is met by framing pipeline design as an objective-driven trade-off between sensitivity/track quality and computational cost, using ablations and parameter sweeps to identify design points. This aligns with Week 01 (optimal control framing), Weeks 10–11 (optimisation/MPC), and the unit's focus on formal objective definition. Additionally, track association is an optimisation problem.

A2.1.5 Coverage statement

The thesis addresses LO1–LO4 through critical review, implementation of estimation/decision algorithms, comparative evaluation under constraints, and objective-driven trade-off analysis, demonstrating at least 75% coverage.

A2.2 Case Study 2: ELEC3305 Digital Signal Processing

A2.2.1 LO1. *Describe signals using mathematical concepts such as convolutions, transforms, spectral analyses, linear difference equations, filters, correlation and covariance*

This learning objective is addressed through frequency-domain radar processing. Convolutions will be used in the form of matched filters in the dechirping process. The Fourier transform will be applied with the FFT algorithm over both fast time and slow time to produce range and Doppler maps. Hann, Hamming, and other filters will be applied to minimise spectral leakage, and the spectral effect of our hardware configuration and geometry will be compensated for. Correlation, linear difference, and covariance will be essential concepts in the process of detection and tracking of targets. This aligns strongly with Week 03 (convolution and frequency response), Weeks 07–10 (DFT/FFT properties and spectral analysis), and the unit's emphasis on transform-based signal description.

A2.2.2 LO2. *Use software, such as MATLAB and Python, to develop code to analyse and process signals including filtering, inverse filtering, spectral analyses, resampling, time-frequency analyses*

This learning objective is addressed through substantial software implementation and testing of DSP stages in an end-to-end pipeline, including benchmarking CPU vs GPU implementations and developing validation/profiling tools with parameter sweeps. This aligns with Weeks 02–12 practicals, which focus on implementing DSP in software. Python and C++ will be used to develop both the pipeline and the simulation application.

A2.2.3 LO3. *Identify linear signal models for time-invariant systems such as convolutional models, sinusoidal and harmonic models, echo and reverberation models, modulation models and time-frequency models.*

This learning objective is addressed by using standard linear and time-frequency models for radar returns and mapping sampled signals into range–Doppler representations, then studying how model behaviour changes under low SNR and limited resolution. This aligns with Week 02 (signal models), Weeks 04–06 (DTFT/Z-transform/transfer functions), and Weeks 11–12 (sampling and ADC/DAC considerations).

A2.2.4 LO4. *Design and evaluate signal processing systems.*

This learning objective is addressed directly by designing and evaluating alternative pipeline configurations (windowing, FFT sizing/batching, CFAR variants, integration policies, and angle sensing), and quantifying effects on detection sensitivity, false alarms, and end-to-end latency/throughput. This aligns with Weeks 04–10 and the unit's system design emphasis.

A2.2.5 Coverage statement

The thesis covers LO1–LO4 through transform-based processing, software implementation, linear/time-frequency modelling, and end-to-end system design and evaluation, demonstrating at least 75% coverage.

A3 Work Health and Safety Report

TODO: Write the health and safety report

TODO: I don't quite know what this involves, but it is worth 5% so I shouldn't ignore it

A3.1 Work, Health and Safety at Mission Systems

A3.2 Incidents and Preventions During Placement

A4 Practical Experience Logbook